

Scalability & Deployment Plan

Step 6 · plan for scale to ~80,000 cameras

PROBLEM STATEMENT

Integrated Video Management & Analytics Platform

FOR

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SUBMITTED BY

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SOURCE

docs/scalability-plan.md

Answers Step 6 (*Plan for Scale*) and the sizing, cost and deployment areas of Step 3 (*Expected Solution Approach*). Companion to `high-level-design.md`, which carries the architecture itself.

0. The number this plan is built on

Every figure below is driven by one measured quantity rather than an assumed one.

I surveyed all 30 cameras on the government sandbox grid and measured what each can actually resolve. **Three can read a number plate.** The median vehicle presents 27.4 px of plate against the ~120 px a read requires. On the person axis, the closest people yield 2.7–9.8 px between the eyes against a 40 px floor for attempting face matching.

That measurement is the input to this entire plan. It is why the heavy analytics tier is **10 % of cameras, not 100 %**, and every compute, bandwidth, storage and cost figure below follows from it.

A plan sized on the assumption that all 80,000 cameras need continuous heavy inference is not conservative. It is wrong by an order of magnitude, and it prices the programme out of existence.

On "continuously process the CCTV feeds"

The brief asks for a solution that continuously processes the feeds. This design does — with one qualification that is the whole architecture.

Every onboarded camera is continuously processed **by the analytic its measured capability supports**. A plate-capable camera runs continuous ANPR. A presence-only camera runs continuous detection, tracking and re-identification, and continuous crowd-dispersal monitoring. Nothing is sampled, and no camera is left dark.

What the system does *not* do is run every analytic on every camera. That is not a limitation I accept reluctantly; it is arithmetic. The organisers' own resources page states that each connected client receives its own copy of the stream and instructs entrants to open only the cameras they are actively processing — a constraint that exists in a 30-camera sandbox and does not soften at 80,000. Running plate reading on a camera measured at 27 px of plate does not produce weak reads; it produces confident wrong ones, which cost a control room a dispatch each.

So: continuous on every camera, and *correctly assigned* on every camera.

1. Hardware and software requirements

Software

LAYER	COMPONENT	LICENCE
Ingest	FFmpeg / GStreamer, RTSP-TCP, HLS, ONVIF profile S/T	LGPL / BSD
Inference	PyTorch + torchvision detectors	BSD-3-Clause
OCR	Tesseract 5	Apache-2.0
Vision	OpenCV	Apache-2.0
Service	FastAPI + Uvicorn	MIT / BSD
Registry	PostgreSQL + PostGIS (SQLite in this submission)	PostgreSQL licence
Queue	Redis Streams or Kafka at district tier	BSD / Apache-2.0
Object store	MinIO or department S3-compatible	AGPL / vendor

No AGPL component sits in the processing path. Ultralytics YOLO — the default choice for work of this kind — is AGPL-3.0, which reaches the whole deployed system and is a procurement blocker rather than a footnote. A permissively licensed detector was chosen for that reason.

Hardware, by tier

TIER	SITED AT	ROLE
Edge	Camera site / local NVR room	Decode, detect, track, discard non-events. Sends metadata and evidence clips only.
District	33 district HQs	Aggregation, ANPR on the assigned subset, watchlist correlation, district alerting, 72-hour hot video.
Central	State data centre + DR site	Registry, statewide correlation, long-horizon search, case management, audit chain, warm/cold archive.

The tiering is not for elegance. It is because centralising raw video is impossible on the bandwidth arithmetic in §3, so inference must happen where the video already is.

2. AI processing capacity

The measured unit cost

A datacentre GPU of A100/L40S class sustains on the order of **40–60 simultaneous 1080p streams** at detection workloads in published reference architectures. I use 50 as the planning midpoint and state the range.

This is a vendor-published figure, not one I measured. What I did measure is the relative cost of the two designs, which is what the decision turns on.

The two designs, side by side

	EVERY CAMERA, HEAVY ANALYTICS	CAPABILITY-ASSIGNED
Cameras on heavy inference	80,000	8,000
GPUs required	~1,600	~160
Cameras on cheap presence detection	0	72,000

Presence detection — background subtraction and motion-gated tracking — is CPU work, roughly three orders of magnitude cheaper per camera than continuous detection inference. The 72,000 cameras in that tier are served by district CPU capacity, not by GPUs.

Correction, kept rather than quietly dropped: an earlier draft of my design document put the all-camera figure at 264,000 GPUs, extrapolated from a laptop GPU at 0.33 s per frame. That overstated the requirement by roughly 150×. A number nobody can reproduce is worth less than no number.

Headroom and growth

Sizing assumes the measured 10 % heavy share. The upgrade plan in §8 shows eight cameras on the sandbox grid are within a factor of two of plate capability — fit longer lenses and the heavy tier grows. **Capacity planning should therefore assume the heavy share rises toward 20–25 % as the estate is optimised, not that it stays at 10 %.** That is the correct direction for a programme that is improving its own cameras, and it is why the district tier is specified with GPU expansion slots rather than fixed appliance counts.

3. Network and bandwidth planning

Why centralisation is not available

A 1080p H.264 stream is 2–5 Mbps. Eighty thousand of them is **160–400 Gbps sustained**, statewide, every second of every day. That is not a procurement problem; it is a physical one, and it is the single hardest constraint in the brief.

What actually crosses the network

LINK	CARRIES	ORDER OF MAGNITUDE
Camera → edge	Full stream	existing local cabling
Edge → district	Metadata + evidence clips	2–5 % of raw
District → central	Sightings, alerts, audit entries	kilobytes per event
Central → operator	On-demand pull of a specific clip	bursty, small

Sending metadata rather than video is a 95–98 % reduction. A sighting is a few hundred bytes: camera id, corrected timestamp, tier, confidence, bounding box, and a reference to the clip that remains at the edge until an officer asks for it.

Low-bandwidth and intermittent sites

Rural Panchayat cameras will not have reliable backhaul. The design assumes they do not:

- **Store-and-forward.** Edge nodes buffer metadata locally and reconcile when the link returns. Nothing is lost to an outage; it arrives late and is marked late.
- **Degrade to events, not to nothing.** Below a bandwidth floor the edge stops forwarding thumbnails and forwards only structured sightings.
- **Evidence on demand.** A clip is pulled only when an officer opens that sighting, not pushed on the chance it matters.
- **Clock independence.** Because the registry measures and corrects clock drift (I found a government camera **6 h 12 m wrong**), a late-arriving batch still lands in the correct place on the timeline.

4. Storage and retention strategy

The raw arithmetic

4 Mbps continuous is ~43 GB per camera per day. Across 80,000 cameras that is **3.4 PB per day**, and **103 PB at 30-day retention**. Full retention of raw video statewide is not viable at any plausible budget.

Tiered retention

TIER	HOLDS	RETENTION	WHERE	ACCESS
Hot	Raw video, rolling	72 hours	Edge / district NVMe + HDD	Instant scrub
Warm	Evidence clips attached to a case; alerted sightings	1 year	District object store	Seconds
Cold	Case-bound evidence under legal hold; archived audit chain	7 years, or as the case requires	Central archive / tape	Minutes to hours
Indefinite	Metadata: sightings, alerts, capability measurements, audit entries	No expiry	Central PostgreSQL	Instant, searchable

The inversion is deliberate: **video expires, metadata does not**. A sighting record is a few hundred bytes and remains searchable for years; the video behind it survives 72 hours unless something made it evidence.

Storage volume under this policy

TIER	CALCULATION	VOLUME
Hot	80,000 × 43 GB × 3 days	~10.3 PB rolling, distributed across districts
Warm	assume 0.1 % of sightings become evidence clips	tens of TB per year
Cold	case-bound only	low TB per year
Metadata	~500 bytes × est. 2 bn sightings/year	~1 TB per year

Hot storage is the dominant cost and it is the one that is distributed — roughly 310 TB per district across 33 districts, which is a rack, not a data centre.

Per-department retention

Retention is a property of the owning department, not a global setting. Health footage carries patient confidentiality above ordinary privacy duty. The registry holds a `governance_class` per camera for exactly this, and retention policy binds to it.

Stated plainly: the `governance_class` column exists and is populated; the retention enforcement that acts on it is designed here and not yet built. It is the largest gap between this plan and the running code.

5. Load balancing, horizontal scaling, monitoring

Scaling shape

- **Edge and district tiers scale horizontally by camera count.** A district worker owns a disjoint set of cameras; adding cameras adds workers. There is no shared mutable state between workers, so this scales linearly.
- **Assignment is the load balancer.** The registry already decides which camera gets which analytic; the same decision distributes work across workers by cost class, so a district does not end up with all its plate-capable cameras on one node.
- **The central tier scales read-heavy**, with PostgreSQL read replicas for search and a single writer for the audit chain. The chain must remain serialised — a forked hash chain reports as tampered for the rest of its life — so audit writes are deliberately not distributed.

Monitoring, logging, health checks

The registry already measures camera condition; monitoring extends that rather than bolting on a separate system.

SIGNAL	SOURCE	ACTS ON
Camera reachable, delivering frames	Ingest worker	Registry <code>last_seen_live</code>
Delivered vs declared frame rate	PTS measurement	Health score; alerts on divergence
PTS gaps, decoder errors	Ingest worker	Stream quality
Capability drift	Periodic re-survey	Alerts if a camera's measured plate width falls — it has been knocked, refocused or obscured
Inference queue depth, GPU utilisation	District workers	Autoscale / alert
Audit chain integrity	Scheduled verification	Security alert on break

Capability drift is the one worth emphasising. A camera that was plate-capable and is now presence-only has physically moved or been obstructed. Conventional monitoring reports it as healthy — it is online and delivering frames. Only a system that measures capability notices.

6. High availability, backup, disaster recovery

CONCERN	APPROACH	TARGET
Edge node failure	Cameras re-home to a neighbouring node; degraded coverage, not lost coverage	< 5 min
District failure	Standby district takes assigned cameras; hot video for that district is lost, metadata is not	RTO < 1 h, RPO ~0 for metadata
Central failure	Warm standby at a second site; streaming replication of the registry	RTO < 4 h, RPO < 5 min
Network partition	Districts operate autonomously and reconcile on restore	Degraded, not down
Data loss	Registry backed up continuously; audit chain replicated and independently verifiable	RPO < 5 min

The audit chain is the one thing that must never be silently restored. A backup restore that rewinds the chain creates a gap that is indistinguishable from tampering. So chain segments are replicated append-only and verified after any restore, and a verification failure is an incident rather than a log line.

Graceful degradation, in order of what is given up: 1. Face/person analytics (already unassigned on this grid) 2. Cross-camera re-identification 3. ANPR on occasional-yield cameras 4. Presence detection 5.

Last to go: alerting on confirmed plate matches from plate-capable cameras, because that is the capability the programme exists for.

7. Cybersecurity architecture

CONTROL	IMPLEMENTATION	STATUS
Transport	TLS on every hop; RTSP forced over TCP	Built
Credential handling	Environment-only, never in source or config committed to VCS	Built
Case-linked access	No movement query answered without case ref + searching officer + authorising officer, refused before a row is read	Built and tested
Tamper-evident audit	SHA-256 chain; any edit breaks every subsequent hash	Built and tested
Role-based access control	Officer / supervisor / auditor / administrator, with query scope and export rights bound to role	Designed, not built
Authentication	Integration with departmental identity (eGujCop / AD), MFA for export and watchlist edit	Designed, not built
Network segmentation	Camera VLANs isolated from corporate; district tier in a DMZ	Design
Masking on export	Uninvolved parties obscured in evidence bundles	Designed, not built
Data minimisation	Metadata retained, video expired; see §4	Partly built

Stated without softening: there is currently no authentication and no RBAC. The case-linked gate is real, tested, and refuses correctly — but it trusts the officer name it is given. In deployment that gate must sit behind departmental identity, and the roles above must bind to it. Shipping the gate without the identity behind it would be security theatre, and I would rather name it here than let a reviewer discover it.

8. Estimated implementation and operational costs

These are order-of-magnitude planning figures for comparing designs, not a tender response.

Hardware pricing varies with procurement route, volume and year; every figure below should be validated against current GeM rates before any commitment. They are included because the brief asks for them and because the *ratio* between the two designs is robust even where the absolute numbers are not.

Assumptions, stated per line so each can be disputed separately. One *node* is one datacentre-class GPU plus its host server: an A100/L40S-class card is roughly ₹8–13 lakh and the server around ₹8–12 lakh, so **₹20 lakh per node** is the midpoint of a defensible range. Continuous draw 700 W per node (GPU ~400 W, host ~300 W) at ₹8/kWh. Hot storage at **₹12,000 per TB installed** — redundancy, chassis and controllers included, not the price of a bare drive. District tier reuses existing department server rooms.

ITEM	EVERY-CAMERA DESIGN	CAPABILITY- ASSIGNED	NOTE
GPU nodes	~1,600	~160	Driven by the measured 10 % share
GPU capital	₹320 cr	₹32 cr	at ₹20 lakh/node
GPU power, per year	₹7.8 cr	₹0.8 cr	700 W continuous
Hot storage, 10.3 PB	₹12.4 cr	₹12.4 cr	identical — a function of camera count, not of analytic
Network	statewide 160–400 Gbps backhaul	district-local only	the larger hidden cost, not priced here
Indicative capital	~₹332 cr	~₹44 cr	

The difference is ~₹288 crore of capital, and roughly ₹7 crore a year in power, and it rests on one measurement that took a day: surveying 30 cameras and finding that three of them can read a plate.

Storage is deliberately shown as identical in both columns. Hot video depends on how many cameras exist, not on which analytic runs over them, so including it in the "saving" would inflate the comparison. The saving is the GPU line, and only the GPU line.

An earlier draft of this table was out by a factor of ten on every row — ₹1,600 cr against ₹160 cr, power at ₹52 cr, storage at ₹41 cr — because the per-unit assumptions and the totals were written at different times and never reconciled. Every one of those errors inflated the saving being claimed. It is corrected here rather than quietly restated, and the figures are now computed in `tools/deck_data.py` from the per-unit constants above, so the deck and this document cannot drift apart again.

The cheapest capability available

The upgrade plan (`tools/plan_upgrades.py`) identifies, per camera, what would have to change for it to do the job:

VERDICT	CAMERAS (OF 30)
Already plate-capable	3
A longer lens or a different aim	8
Move it, or add one closer	8
New camera, new site	4
No vehicles pass it — wrong question	7

Eight cameras are within a factor of two of plate capability. Fitting longer lenses takes the grid from 3 plate-capable cameras to 11, of the 23 that watch a road at all — for the cost of eight lenses rather than eight cameras. Extrapolated statewide, a lens-first optimisation programme is the highest return per rupee available to this deployment, and it is invisible to anyone who has not measured.

9. Statewide rollout plan

PHASE	SCOPE	DURATION	EXIT CRITERION
0 — Proof	~50 sandbox cameras	Complete	Live test, 12–13 Oct
1 — District pilot	One district, all departments, ~2,000 cameras	3 months	Survey complete; assignment stable; alert budget holding
2 — Optimise before expanding	Lens/aim programme on pilot district	2 months	Measured plate-capable share risen; cost per confirmed sighting falling
3 — Regional	8 districts, ~15,000 cameras	6 months	District tier autonomous under partition
4 — Statewide	33 districts, ~80,000 cameras	18 months	Federation across all 26 departments
5 — Steady state	Continuous re-survey	Ongoing	Capability drift detected and corrected

Phase 2 is the unusual one and it is deliberate. Most rollouts expand camera count; this one pauses to improve the cameras already onboarded, because a lens is two orders of magnitude cheaper than a camera and the measurement says which lenses. Expanding before optimising multiplies the number of cameras that cannot answer the question being asked of them.

Department-wise information requirements

To onboard a department I need, per camera, only what the registry cannot measure for itself:

REQUIRED	WHY	IF UNAVAILABLE
Stream endpoint + credentials	To connect at all	Blocking
Owning department	Governance class, retention, access	Blocking
Site location (lat/lon or address)	GIS, route reconstruction, look-ahead	Degrades to non-spatial
Contact for physical access	Lens/aim changes in phases 2	Blocks optimisation only
VMS make/model + API or ONVIF profile	Federation adapter selection	Falls back to RTSP/HLS
Existing retention policy	Reconciling mine with theirs	Assume department default

Deliberately **not** required: resolution, frame rate, codec, field of view, mounting height, focal length, or any capability claim. The system measures all of those from the video, because on the sandbox grid the catalogue's own declared values were wrong — one camera declares 250 fps in its container header while reporting 29.96 in another field of the same file. Asking 26 departments for specification data that turns out to be unreliable would delay onboarding and then mislead the assignment layer.

This is the single biggest reduction in onboarding friction the design offers: a department supplies an endpoint and an owner, and the platform works out the rest.

10. What this plan assumes, and what would change it

ASSUMPTION	IF WRONG
40–60 1080p streams per datacentre GPU	Linear effect on GPU count; ratio between designs unchanged
Measured 10 % plate-capable share holds statewide	Sandbox is 30 cameras. A statewide survey is phase 1's first task, and the share is the one number this plan is most sensitive to
0.1 % of sightings become evidence	Warm storage scales linearly; still small against hot
District server rooms can host the district tier	Otherwise add ~33 small facilities
Backhaul exists to district HQs	Phase 3 gating item; store-and-forward covers gaps, not absence

The honest summary: the architecture is robust to most of these being somewhat wrong, and sensitive to one — the statewide capability distribution. Which is why the first action in phase 1 is to measure it, not to assume that 30 cameras in a sandbox represent 80,000 across 26 departments.